SOHAN SAI VANAM

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EDUCATION

University of Texas at Arlington **Arlington, TX**
Honors Bachelor of Science in Mechanical Engineering *August 2024 – Present*

- **GPA:** 3.662
- **Relevant Coursework:** Mechanical Design, GD&T, Mechanics of Materials, Thermodynamics, Satellite Technology

SKILLS

CAD & Reverse Engineering: SolidWorks, ScanTo3D, Siemens NX, Onshape, point-cloud processing, reverse engineering, CAD modeling.
Engineering Design: GD&T, engineering drawings, tolerance specification, design for assembly, FEA
Prototyping & Testing: 3D printing, mechanical assembly, bench testing, metrology, troubleshooting
Programming & Tools: MATLAB, C, Arduino Uno, Microsoft Excel, Teamcenter, Altair FEKO

EXPERIENCE

Brahmastra Aerospace & Defence Pvt. Ltd. **Arlington, TX**
Project Intern *September 2023*

- Designed and optimized a Yagi-Uda antenna in Altair FEKO by iterating geometry and evaluating performance for satellite communication applications.
- Analyzed radiation patterns, gain, and directivity to compare design iterations and validate simulated antenna performance.

University of Texas at Arlington - CHEM 1465 **Arlington, TX**
Peer-Led Team Learning Leader *August 2025 – July 2026*

- Led two weekly peer learning sessions for separate groups of 12 students using structured, faculty-developed packets.
- Helped students understand fundamental chemistry concepts to improve course performance.

PROJECTS

2-Axis Robotic Arm **Arlington, TX**
MAE 1351 – Engineering Design

- Designed and built a 2-axis robotic arm on a belt-driven linear slide system using SolidWorks, aluminum extrusions, and servo motors, integrating control through Arduino Uno and C programming.
- Created engineering drawings with dimensions, tolerances, and GD&T for fabricated and 3D-printed components.
- Performed FEA on joints and arm components to evaluate stress distribution and factor of safety, and optimized the design.
- Assembled and tested the system, troubleshooting mechanical and control issues, and iterating the design to improve reliability.
- Recognized as the most innovative project of the semester and later showcased as a reference design in subsequent classes.

Point-Cloud Reverse Engineering **Arlington, TX**
MAE 1351 Honors Contract | Spring 2025

- Converted optical-scanner point-cloud data of a real part into a complete 3D solid model using SolidWorks ScanTo3D.
- Processed scanned geometry and reconstructed component features using reverse-engineering and scanner-based metrology techniques.

Two-Wheeled Biped Robot **Arlington, TX**
Personal Project *July 2026 – Present*

- Designing an Ascento-style wheel-legged robot with articulated legs and independently driven wheels for dynamic balancing and variable-height locomotion.
- Designed a compact cycloidal joint actuator in SolidWorks, considering torque, backlash, weight, bearing support, and manufacturability.
- Bench-testing the actuator and developing the remaining mechanical structure and LQR-based stabilization controller before complete system integration.

OTHER EXPERIENCE

Entrepreneurship Club **Arlington, TX**
Head of Program Coordinators *January 2025 – Present*

- Led a team of 4 program coordinators to plan workshops, speaker events, and student-focused initiatives.

UTA Formula SAE **Arlington, TX**
Suspension Team Volunteer *July 2026 – Present*

- Joined the Suspension Team for the 2026-27 Formula SAE vehicle and began supporting work involving suspension geometry, load paths, component design, and fabrication.

CERTIFICATIONS

Certified SolidWorks Design Professional (CSWP) – Dassault Systèmes **Arlington, TX**
Issued: Aug 2026 | Certificate No.: C-DC5A832N6R